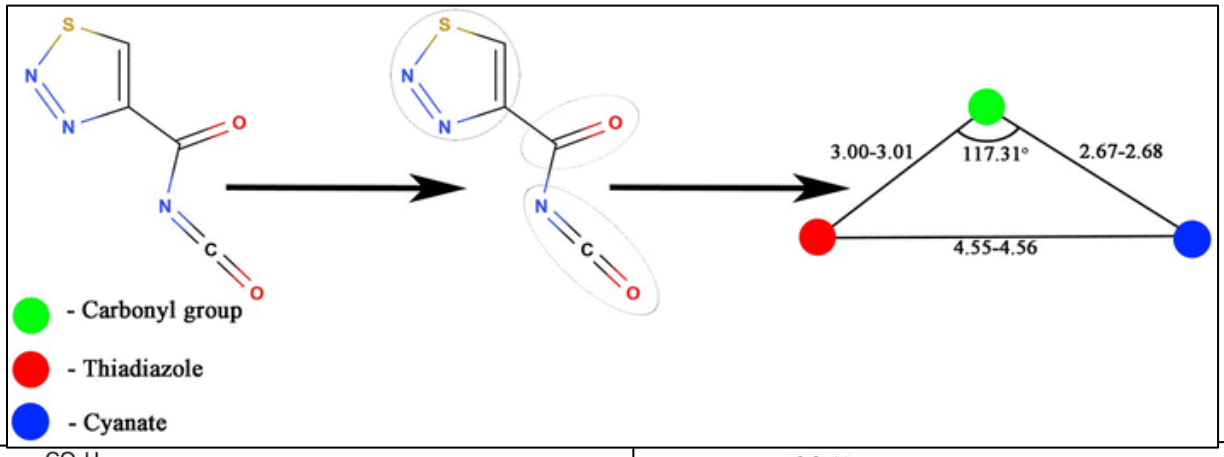
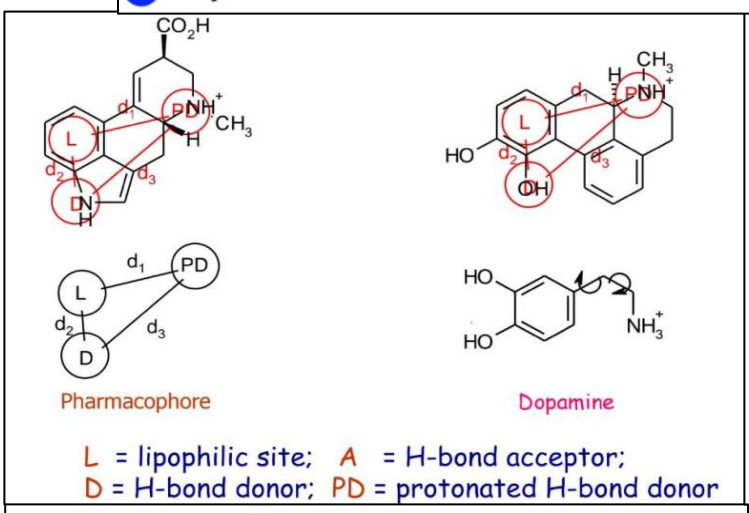
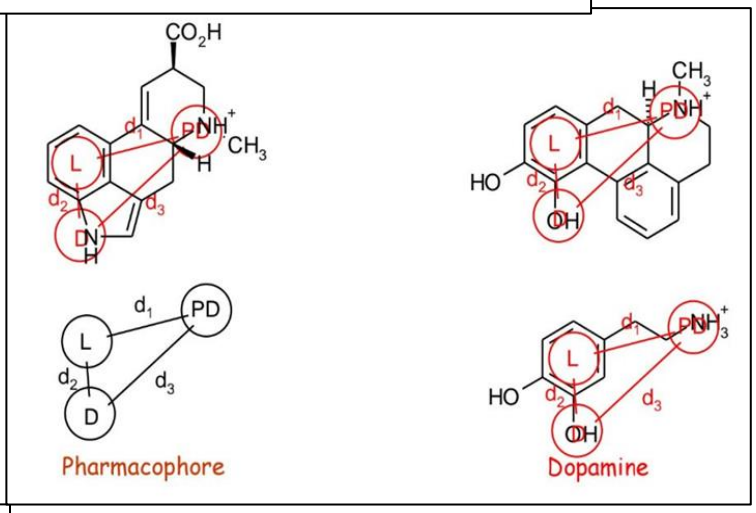
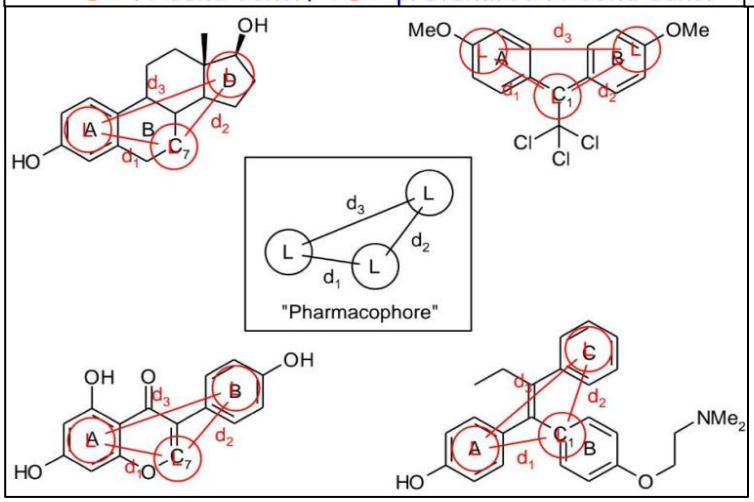
Pharmacophore Selection

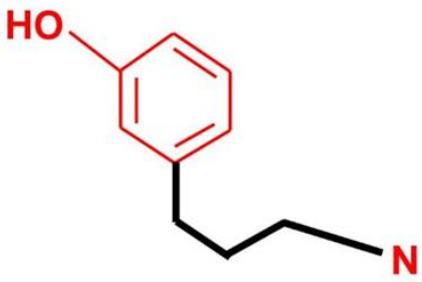


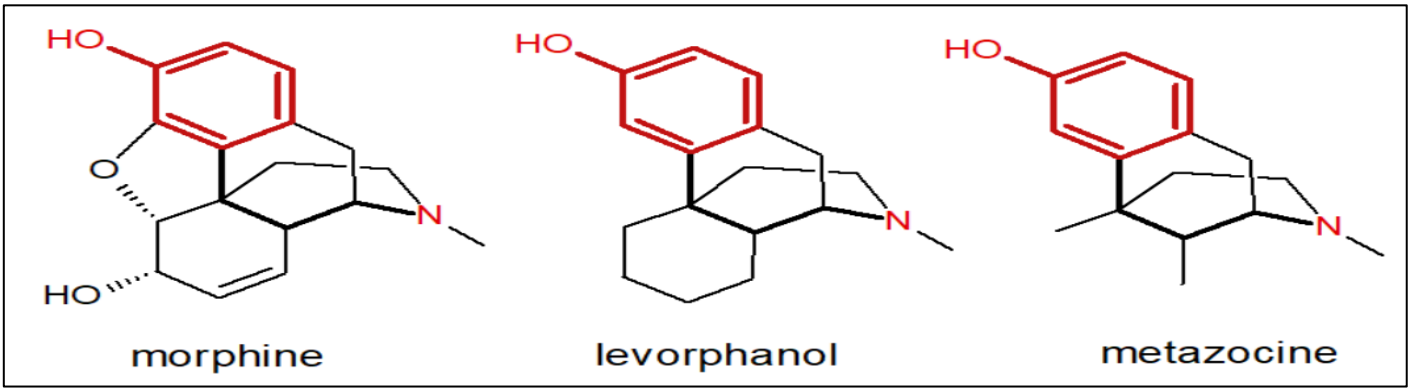


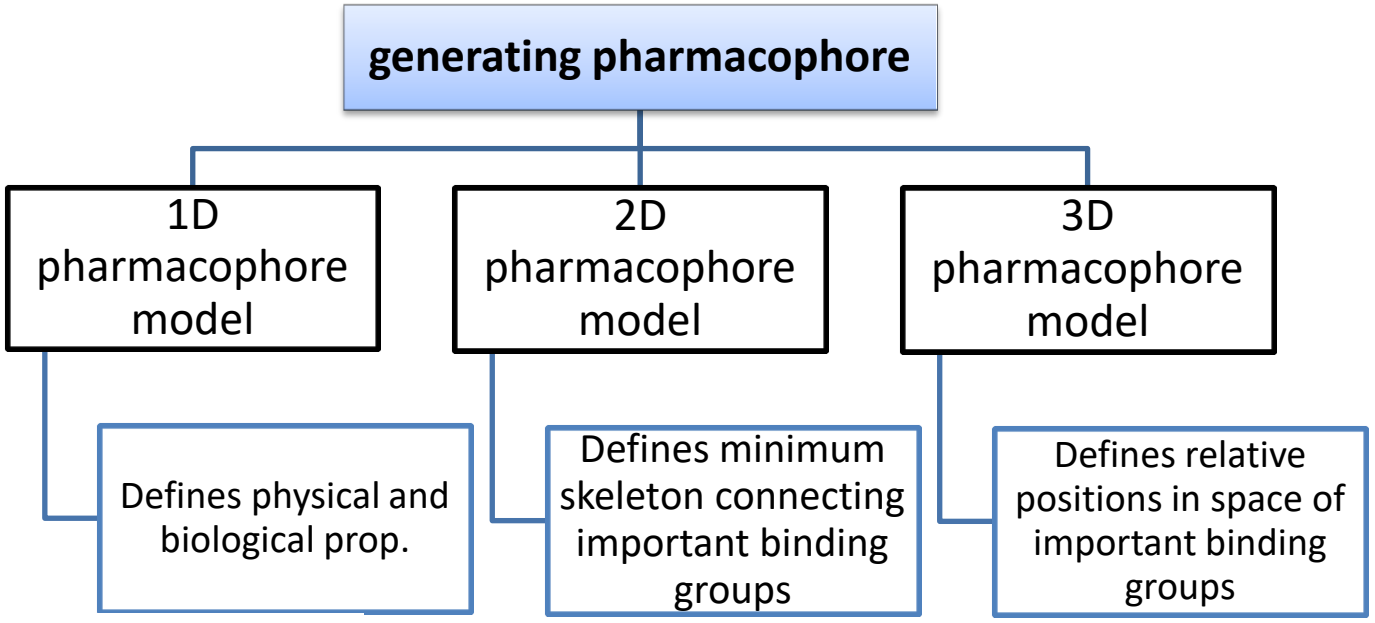


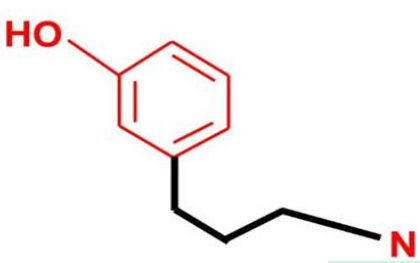


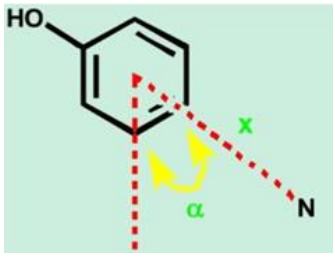
Opiates For Pharmacophore

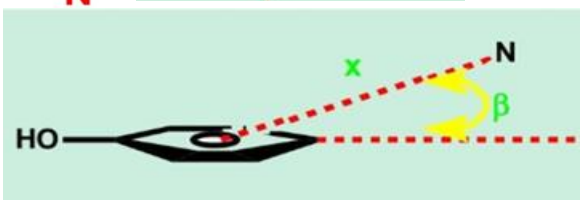


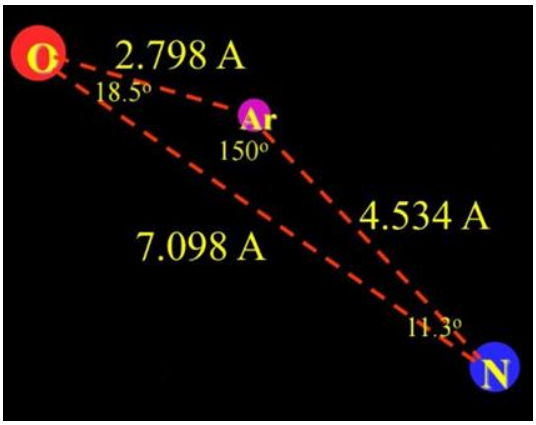






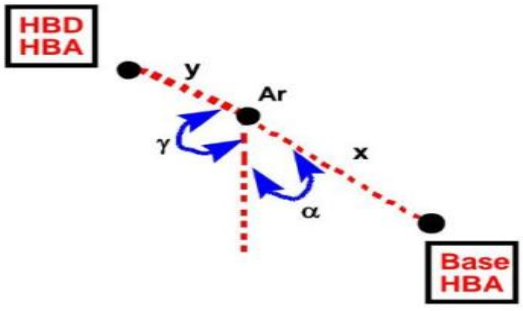


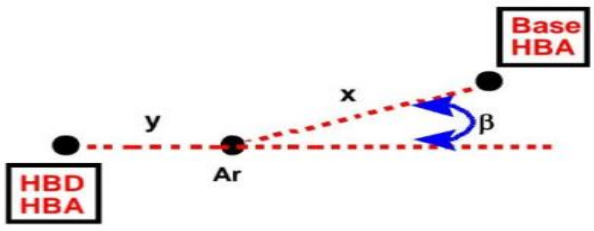




Generalized Bonding Type Pharmacophore

Defines relative positions in space of the binding interactions which are required for activity / binding





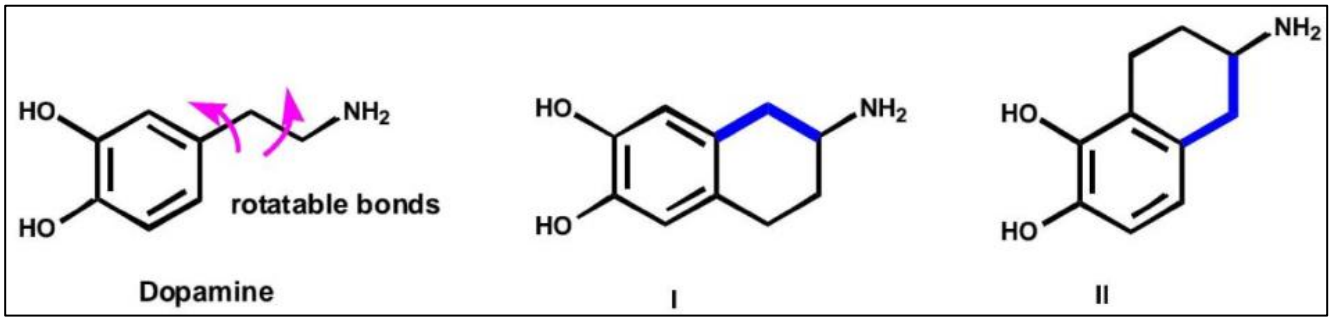
The Active Conformation

• Need to identify the active conformation in order to identify the 3D pharmacophore

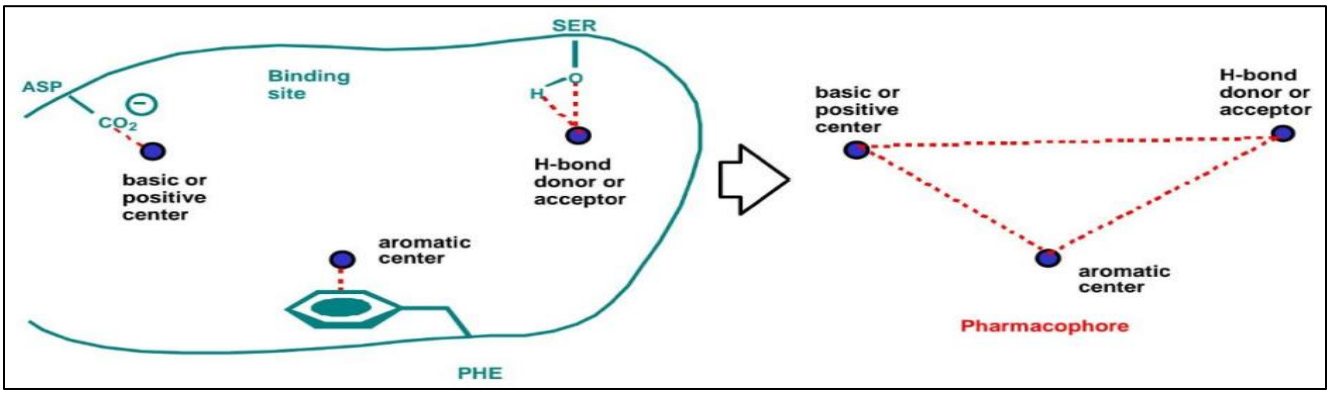
• Conformational analysis - identifies possible conformations and their activities

• Conformational analysis is difficult for simple flexible molecules with large numbers of conformations

• Compare activity of rigid analogues



Pharmacophores from Target Binding Sites



Me-too Drugs

Me-too drugs can be broadly defined as chemically related to the prototype, or other chemical compounds which have an identical mechanism of action.

Benefits

• Some follow-on drugs are better than the pioneer drug

• increased choice between drugs / pioneer drug is ineffective or entails undesirable side effects

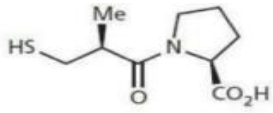
• led to substantial price reductions

Disadvantages:

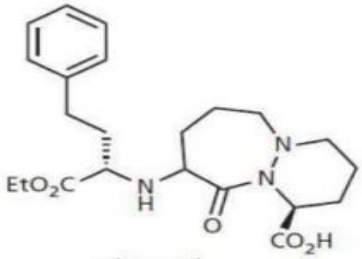
• me-too drugs reduce the incentive to undertake pioneering innovation.

• me-too drugs may have an unacceptable benefit/risk ratio, if their incremental benefits are relatively small.

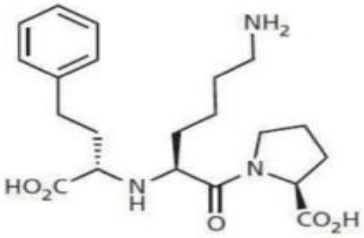
• me-too drugs may be using up more resources than they are worth.



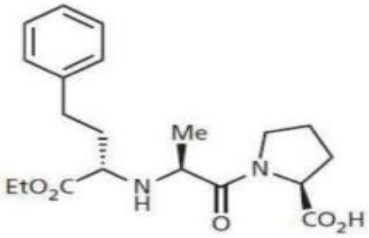
Captopril



Cilazapril
(Hoffmann-LaRoche)



Lisinopril
(Merck)



Enalapril
(Merck)

Examples:

I) Captopril (Anti-hypertension) used as lead compound by different companies to produce their own anti-hypertension drugs.

ii) Modern penicillins are more selective, more potent and more stable than original penicillins

Drug reposition / drug repurposing

is the use of an existing drug for a new treatment that was not indicated before. It has received immense attention during the COVID-19 pandemic emergency.

Example

1. PPI & Anti-malarial drug for COVID-19
2. Aspirin for colorectal-cancer

Advantages:

1. No need for full investigation (cost effective)
2. No need for formulation (reduction in time taken)
3. Skipping preclinical tests
4. Uncovering novel taregets or pathways

De Novo Drug Design

- The design of novel agents based on a knowledge of the target binding site

Procedure:

• Crystallize target protein with bound ligand

• (e.g. enzyme + inhibitor or ligand)

• Acquire structure by X-ray crystallography

• Identify binding site (region where ligand is bound)

• Remove ligand

• Identify potential binding regions in the binding site

• Design a lead compound to interact with the binding site

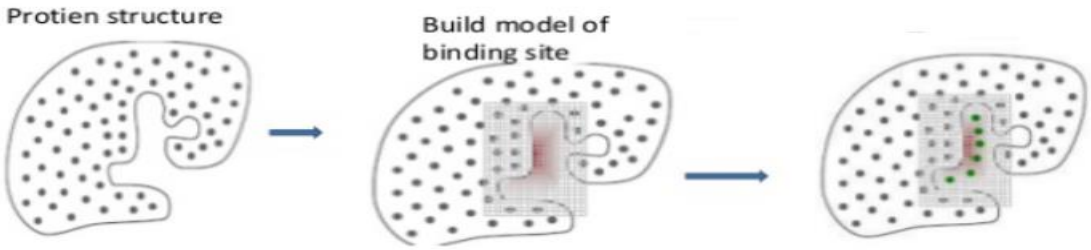
• Synthesize the lead compound and test it for activity

• Crystallize the lead compound with target protein and identify the actual binding interactions

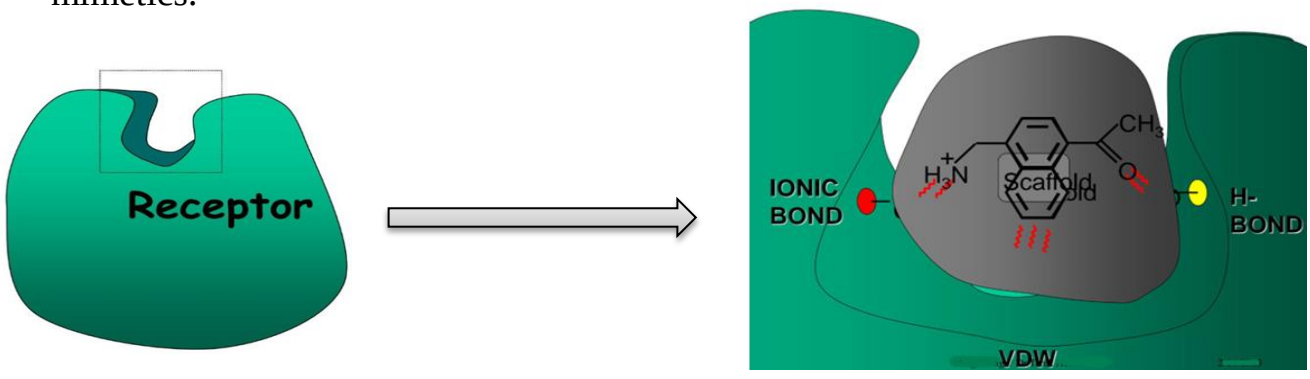
• Structure based drug design **(SBDD)**

Lead Compounds - de novo drug design

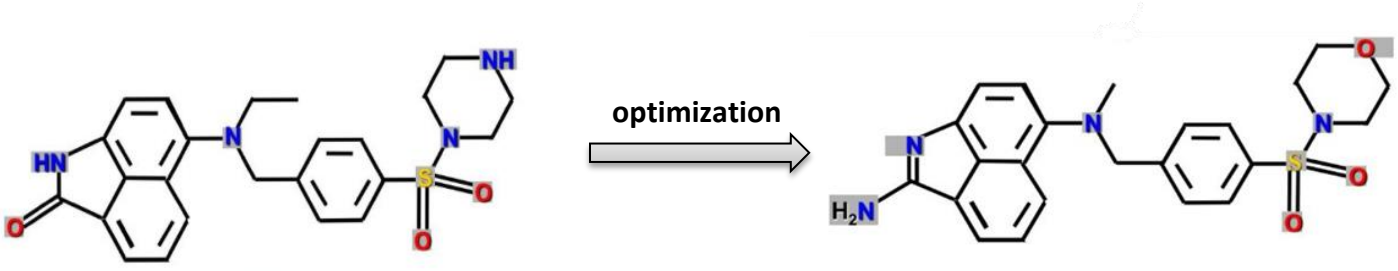
- 3D diagram needs improved
- X-ray crystallography (to identify Protein structure)



Scaffold is a fixed part of a molecule on which functional groups are substituted or exchanged. The protein that is used as a starting point for the design of antibody mimetics.



Thymidylate kinase inhibitors: (Anti- Cancer Agent)



Lead compound prototype

Computer-Assisted Drug Design (CADD)

In practice molecular modeling uses two approaches:

1. direct design of active substances (SBDD) which can be envisaged when the 3D structure of the target molecule is known. In this case the macromolecule can be built with the aid of computer then the fit of the host molecule with its receptor can be optimized. The structure of the ligand, its substituents and confirmation can be modified and the most favorable conditions for interaction (docking) can be simulated on the screen.

gDock: Web Docking Tool

Sketch Structure ⟶ Docked Structure

2. Indirect design of active substances (LBDD), on the other hand, constitutes the only possible approach when the 3D structure of the target molecule is unknown. In this situation comparison of a set of ligands selective for a given receptor is undertaken in order to reveal the molecular information that the compounds have in common despite apparently different chemical formula.

